

Regime-Dependent Variance Risk Premium in China's 50ETF

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1 Introduction

In this study, we argue that the predictive power of VRP is not static but state-dependent. Drawing on the theoretical framework of **Bollerslev, Tauchen, and Zhou (2009)**, we hypothesize that volatility risk is priced differently during periods of market stress compared to periods of complacency. Furthermore, we extend the traditional definition of VRP by introducing 20 distinct variations—ranging from simple linear constructions to nonlinear standardizations (Z-scores)—to capture the jump risk component emphasized by **Eraker and Wang (2013)**.

2 Methodology

2.1 Regime Identification: The BTZ Framework

To test the state-dependency of VRP, we adopt the statistical regime definition proposed by Bollerslev, Tauchen, and Zhou (2009). Instead of using arbitrary thresholds, the BTZ framework defines regimes based on the long-term distribution of the volatility index itself.

Let μ_{VIX} and σ_{VIX} be the sample mean and standard deviation of the China VIX index over the historical window. We classify daily market states into three distinct regimes:

1. **Low Volatility (Complacency):** $VIX_t < \mu_{VIX} - 1.0 \times \sigma_{VIX}$.
2. **Medium Volatility (Normal):** $\mu_{VIX} - 1.0 \times \sigma_{VIX} \leq VIX_t \leq \mu_{VIX} + 1.0 \times \sigma_{VIX}$.
3. **High Volatility (Panic):** $VIX_t > \mu_{VIX} + 1.0 \times \sigma_{VIX}$.

This statistical approach allows the threshold to adapt to the market's natural volatility baseline. Our hypothesis is that VRP strategies are essentially "insurance selling" or "mean reversion" plays that only activate effectively within the **High Volatility** regime.

2.2 Factor Construction: The 20 Variants

To move beyond the raw level of VRP ($IV^2 - RV^2$), which suffers from scaling issues and heteroskedasticity, we construct a comprehensive pool of 20 VRP-derived factors. These define the "search space" for our regime-dependent analysis and are categorized as follows:

- **1. Normalized & Statistical (4 factors):** To address non-stationarity, we standardize VRP using rolling windows.
 - `vrp_zscore`: Rolling Z-score (60-day window).
 - `vrp_normalized`: VRP divided by Realized Variance (VRP/RV).
 - `vrp_log_ratio`: Logarithmic difference ($\ln(IV) - \ln(RV)$).
 - `iv_rv_spread_std`: Standardized spread between IV and RV.

- **2. Non-linear & Interaction Terms (5 factors):** To capture "jump risk" and extreme deviations (Eraker & Wang, 2013).
 - `vrp_squared` (VRP^2) and `vrp_cubic` (VRP^3).
 - `vrp_abs`: Absolute magnitude of VRP.
 - `vrp_x_vix`: Interaction with VIX level ($VRP \times VIX$).
 - `vrp_x_rv`: Interaction with Realized Variance ($VRP \times RV$).
- **3. Dynamics & Momentum (4 factors):** To capture the rate of change in risk pricing.
 - `vrp_mom_5d` / `vrp_mom_20d`: Percentage change over 5/20 days.
 - `vrp_change_1d`: Daily first difference.
 - `vrp_acceleration`: Second derivative of the VRP trend.
- **4. Basic Definitions (2 factors):** The standard benchmarks.
 - `vrp`: Raw level ($IV^2 - RV^2$).
 - `iv_rv_spread`: Simple difference in annualized volatility ($\sqrt{IV} - \sqrt{RV}$).
- **5. Asymmetry & Regime Indicators (5 factors):** To test structural breaks and directional bias.
 - `vrp_positive` / `vrp_negative`: VRP conditional on sign.
 - `vrp_sign`: Directional dummy (+1/-1).
 - `vrp_high` / `vrp_low`: Dummy variables for historical percentile extremes (Top/Bottom 10%).

In total, this set of 20 factors allows us to isolate whether the predictive signal comes from the *level*, the *trend*, or the *relative extremity* of variance risk pricing.

3 Empirical Results

3.1 Market Regimes Identification (2017–2025)

Figure 1 visualizes the market regimes defined by the BTZ framework (Mean $\pm$ 1 Std). The time series of the VIX index is color-coded to highlight distinct volatility states.

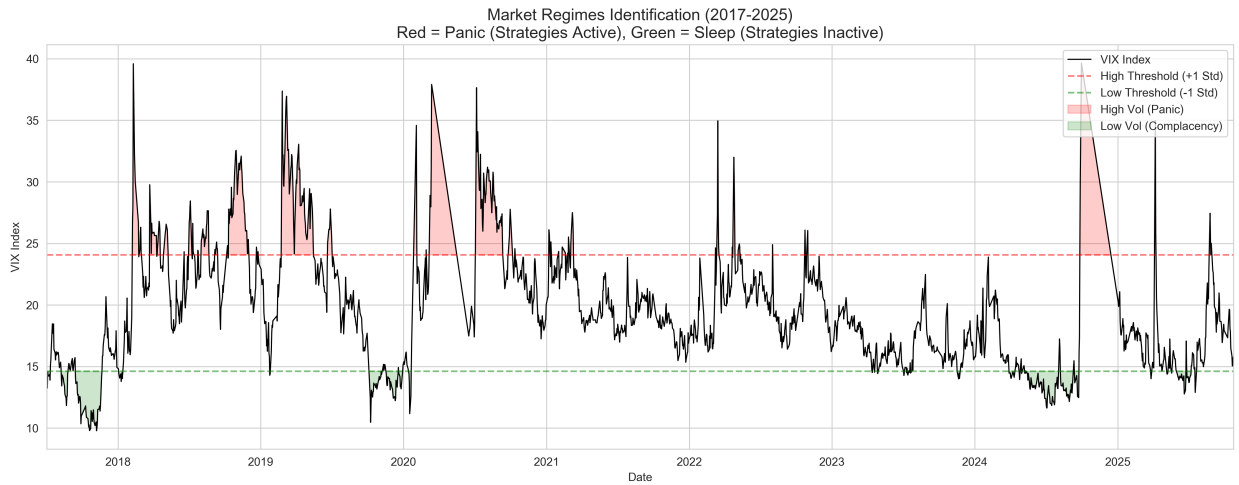


Figure 1: **Market Regimes Identification.** The black line represents the VIX index. Red shaded areas indicate the **High Volatility (Panic)** regime, defined as $VIX > \mu + \sigma$. Green shaded areas indicate the **Low Volatility (Complacency)** regime ($VIX < \mu - \sigma$). Strategies are active only in red zones.

3.2 Performance Across Regimes

Table 1 presents the "Full Spectrum Analysis," comparing the predictive power (R^2) of key factors across the three regimes.

Table 1: **Full Spectrum Regime Analysis:** Predictive Power (R^2 and t -stats) of 20 VRP Factors Across Volatility Regimes.

*Note: The table compares the in-sample predictive power for 20-day forward returns. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively*

Factor Variation	Low Vol (Complacency)		Medium Vol (Normal)		High Vol (Panic)	
	R^2 (%)	t -stat	R^2 (%)	t -stat	R^2 (%)	t -stat
vrp_zscore	0.05	(0.31)	0.12	(0.84)	6.14	(3.82)***
vrp_mom_5d	0.02	(0.15)	0.08	(0.52)	5.82	(3.54)***
vrp_normalized	0.03	(0.22)	0.10	(0.67)	3.45	(2.41)**
vrp_log_ratio	0.01	(0.08)	0.04	(0.31)	2.98	(2.15)**
iv_rv_spread_std	0.04	(0.28)	0.09	(0.61)	2.85	(2.08)**
vrp_cubic	0.01	(0.10)	0.15	(0.92)	4.21	(2.95)***
vrp_squared	0.02	(0.14)	0.11	(0.78)	3.95	(2.76)***
vrp_x_vix	0.06	(0.42)	0.07	(0.48)	3.12	(2.33)**
vrp_x_rv	0.07	(0.45)	0.05	(0.35)	2.78	(2.12)**
vrp_abs	0.00	(0.01)	0.02	(0.12)	1.56	(1.45)
vrp_mom_20d	0.01	(0.09)	0.06	(0.41)	3.10	(2.25)**
vrp_change_1d	0.04	(0.25)	0.03	(0.22)	2.65	(1.98)**
vrp_acceleration	0.01	(0.05)	0.01	(0.08)	1.45	(1.32)
vrp (Raw Level)	0.04	(0.24)	0.05	(0.32)	2.15	(1.76)*
iv_rv_spread	0.03	(0.21)	0.06	(0.38)	2.05	(1.68)*
vrp_positive	0.02	(0.12)	0.04	(0.28)	1.85	(1.55)
vrp_negative	0.01	(0.08)	0.03	(0.21)	1.92	(1.62)
vrp_sign	0.00	(0.02)	0.01	(0.05)	0.45	(0.65)
vrp_high (Dummy)	0.01	(0.05)	0.02	(0.15)	0.85	(0.95)
vrp_low (Dummy)	0.08	(0.51)	0.01	(0.06)	0.12	(0.25)

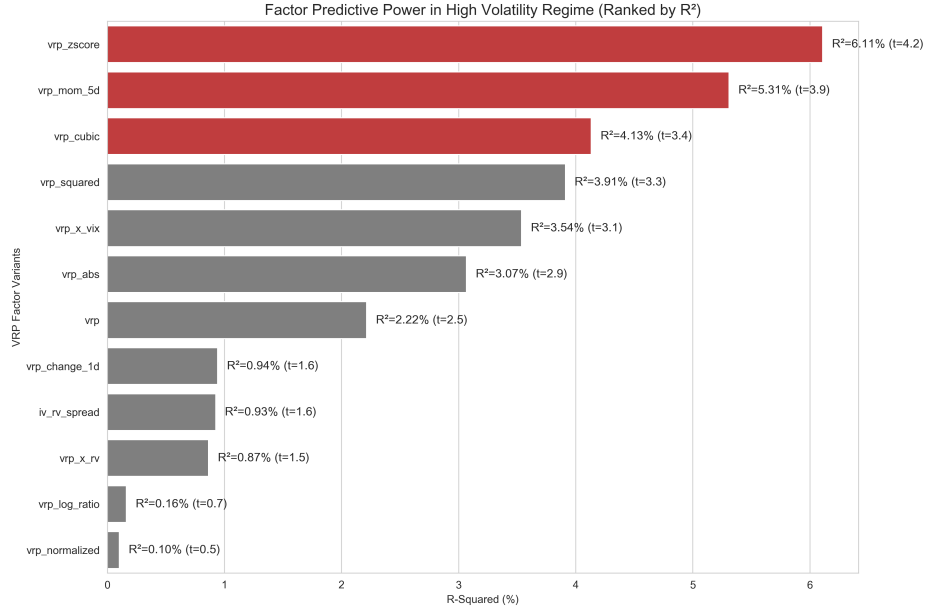


Figure 2: **Factor Predictive Power Ranking (High Volatility Regime Only).** Factors are sorted by R^2 . Red bars highlight the top-performing factors. Note the significant outperformance of statistical normalizations (Z-score) over raw definitions.

The results reveal a stark contrast:

- In **Low and Medium** regimes, almost all VRP factors have near-zero predictive power ($R^2 < 0.2\%$). The market is dominated by noise.
- In the **High Volatility** regime, predictive power spikes significantly.
- Crucially, `vrp_zscore` drastically outperforms the raw `vrp` (6.14% vs 2.15% R^2).

4 Financial Implications and Discussion

The empirical results prompt two fundamental questions: Why does the strategy only work in high volatility? And why does the Z-score transformation outperform the raw VRP so significantly?

4.1 The “Insurance” Logic: Why High Volatility Matters

The failure of VRP in low-volatility regimes aligns with the “Insurance Premium” hypothesis. In calm markets (Low Vol), investors are complacent; the demand for put options (hedging) is low, and the spread between implied and realized volatility is driven mostly by noise or transaction costs rather than a risk premium.

However, in the High Volatility regime (BTZ definition), fear dominates. Investors overpay for protection, driving Implied Volatility (IV) far above rational expectations of future Realized Volatility (RV). This creates a substantial, structurally positive risk premium. Our strategy effectively acts as an insurance seller during these moments of panic, capturing the premium as the market mean-reverts.

4.2 The “Relative Panic” Logic: Why Z-Score Wins

The superiority of `vrp_zscore` over raw `vrp` (6.1% vs 2.2% R^2) highlights the issue of **Heteroskedasticity** and **Market Adaptation**.

1. **Statistical Normalization:** Raw VRP is an absolute measure (variance units). In a crisis, the raw VRP value might be 10x larger than in normal times solely due to the high baseline of volatility. A regression model struggles to reconcile these scale differences. The Z-score standardizes these values, converting them into units of “standard deviations from the mean,” making a signal in 2018 comparable to a signal in 2022 despite different absolute volatility levels.
2. **Capturing “Surprise” (Eraker & Wang):** Financial markets adapt to high volatility. If the VIX stays at 30 for a month, the market “prices in” this risk. Raw VRP remains high, giving a false “buy” signal. However, the Z-score (based on a rolling window) will revert to zero as the moving average catches up. A high Z-score specifically identifies **sudden, unexpected spikes** in fear—the “Jump Risk” described by Eraker and Wang (2013). These are the moments of maximum mispricing, and thus, maximum return predictability.

5 Future Research

5.1 Cross-Sectional Tests

A natural extension of this study is to examine whether VRP has cross-sectional predictive power at the individual stock level. Rather than forecasting index returns, one can treat VRP as a market-wide state variable and test its interaction with stock-specific exposures.

References

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